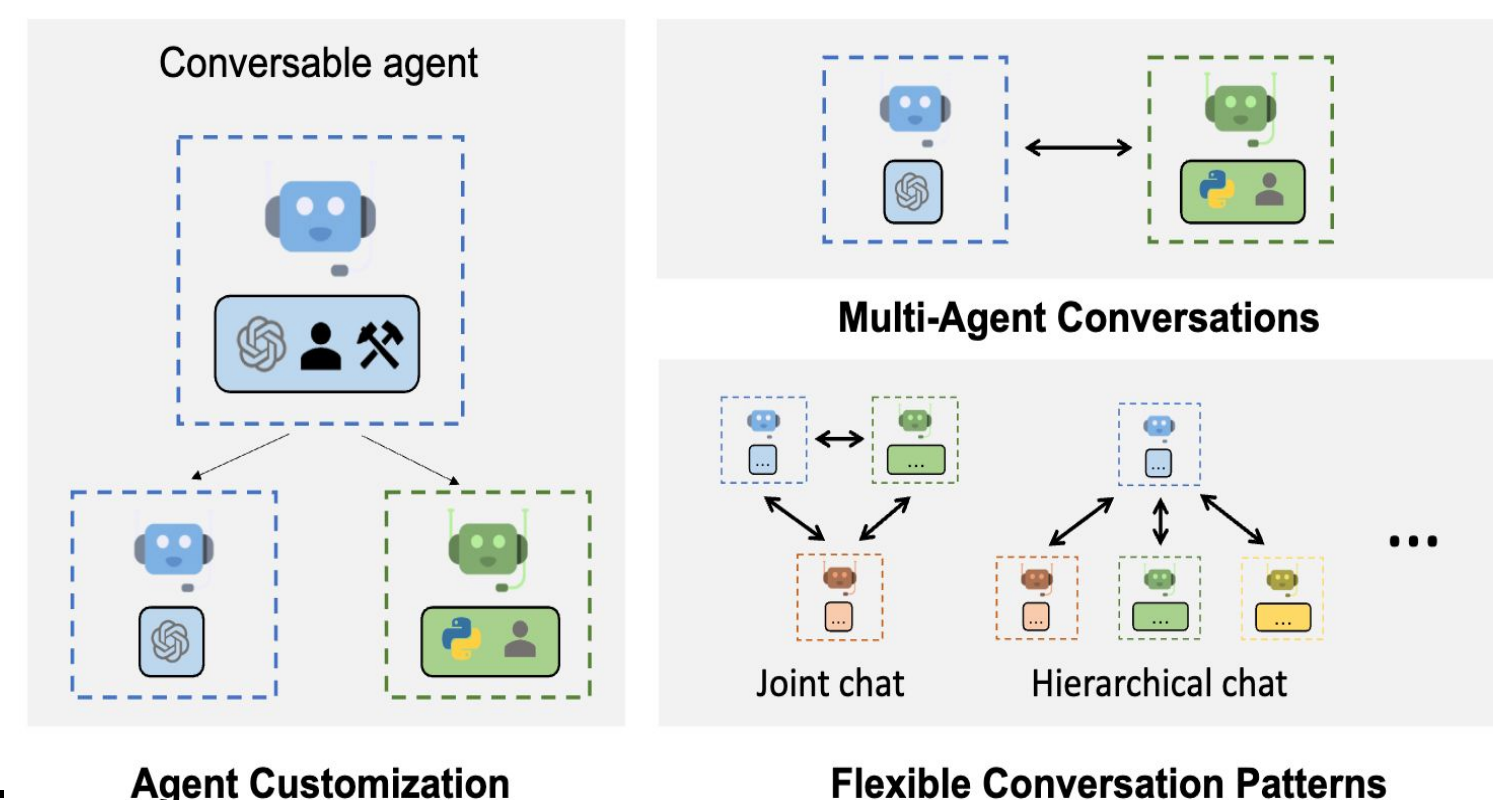


Multi-Agent Large Language Models with AutoGen

Janik Jehkul, Nikita Kostin, Xiangyu Ning

Motivation

- ❖ **Agent:** an entity acting on behalf of human intent, capable of **conversation** and **interaction** with other agents.
- ❖ **AutoGen:** a framework for building **multi-agent** LLM applications, supporting **diverse conversation patterns**.

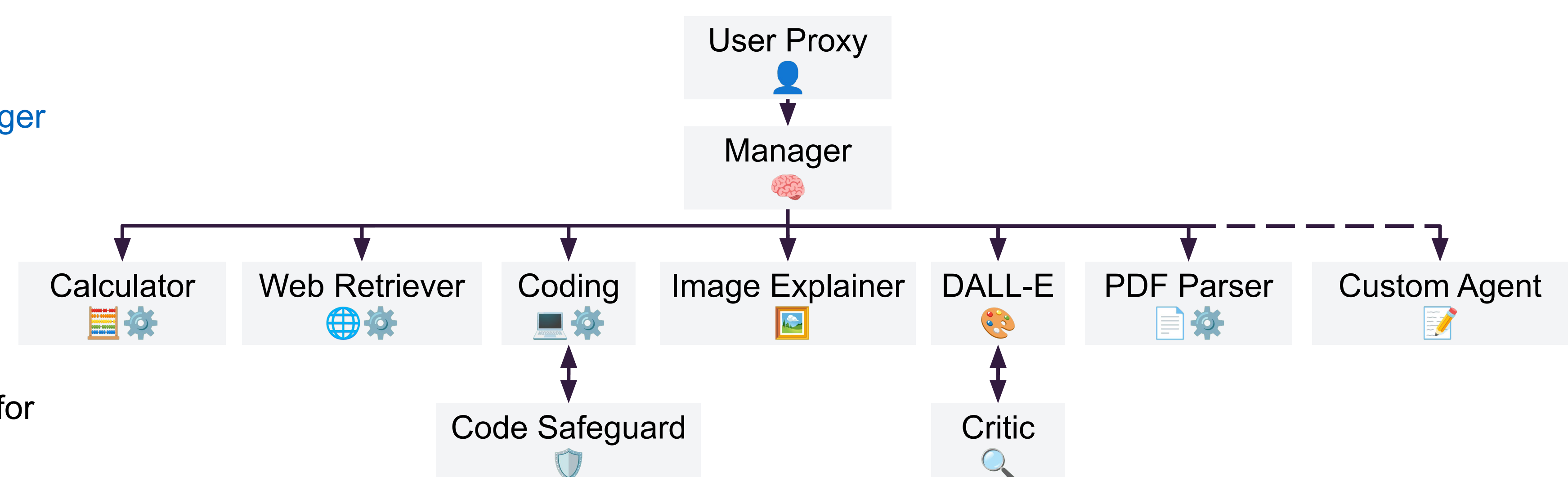


Overview

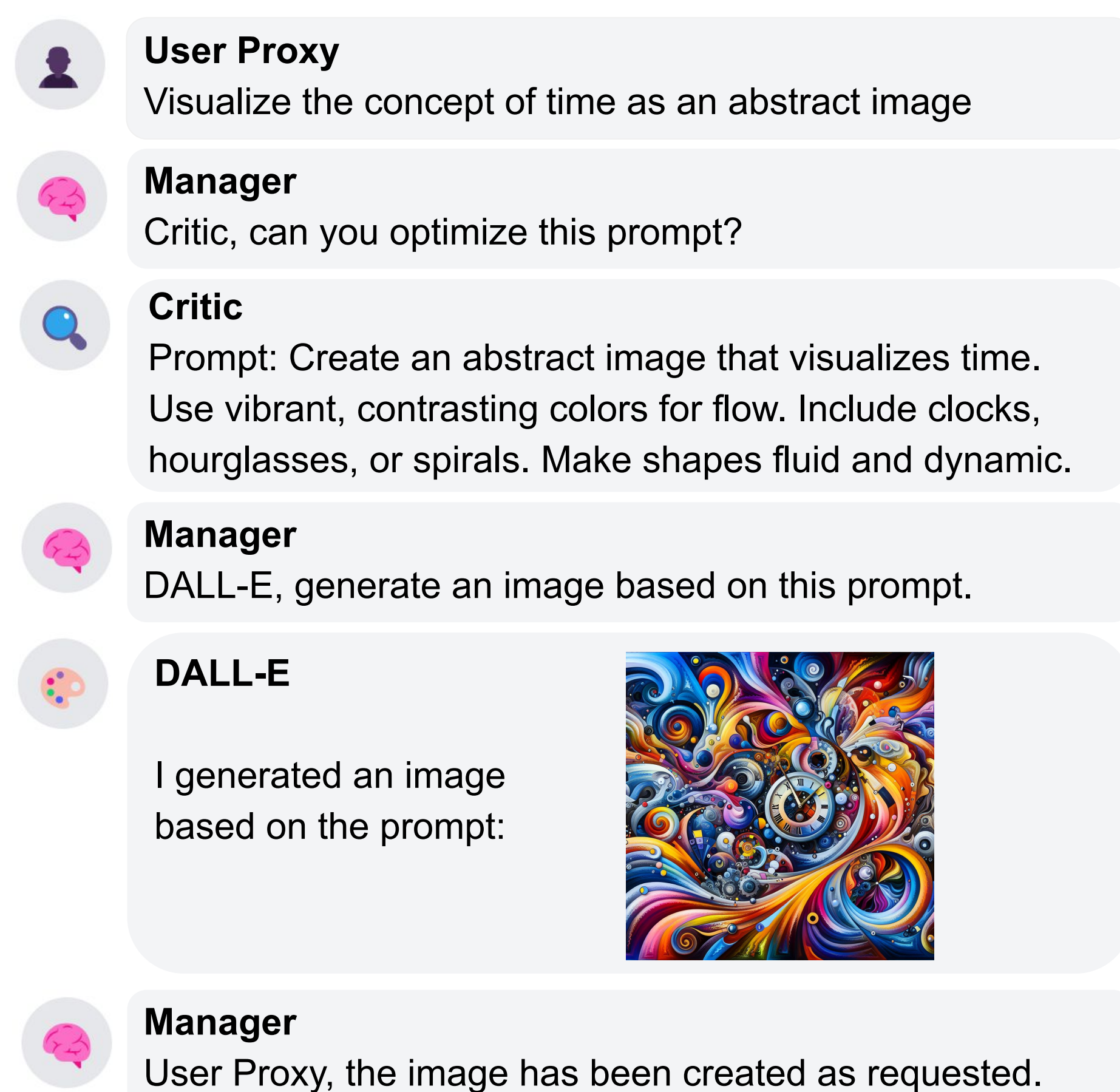
- ❖ Empirical studies demonstrate an advantage of the multi-agent setup over the traditional single-agent approach. An **overarching agent** is required.
- ❖ As a part of this project we:
 - **Reproduced the experiments** and results presented in AutoGen
 - **Explored more scenarios** to showcase the framework's capabilities
 - **Built a parent agent** for various tasks and evaluated its performance on the proposed benchmark
 - **Developed a Streamlit demo** to demonstrate the implemented tasks

Overarching Agent Architecture

- ❖ **User Proxy** passes queries to the **Group Chat Manager**
- ❖ **Specialized agent** subgroups will be selected by **Manager** dynamically for each task
- ❖ **Subgroup agents** have **customized prompts & tools** for their domain (coding, parsing, etc.)

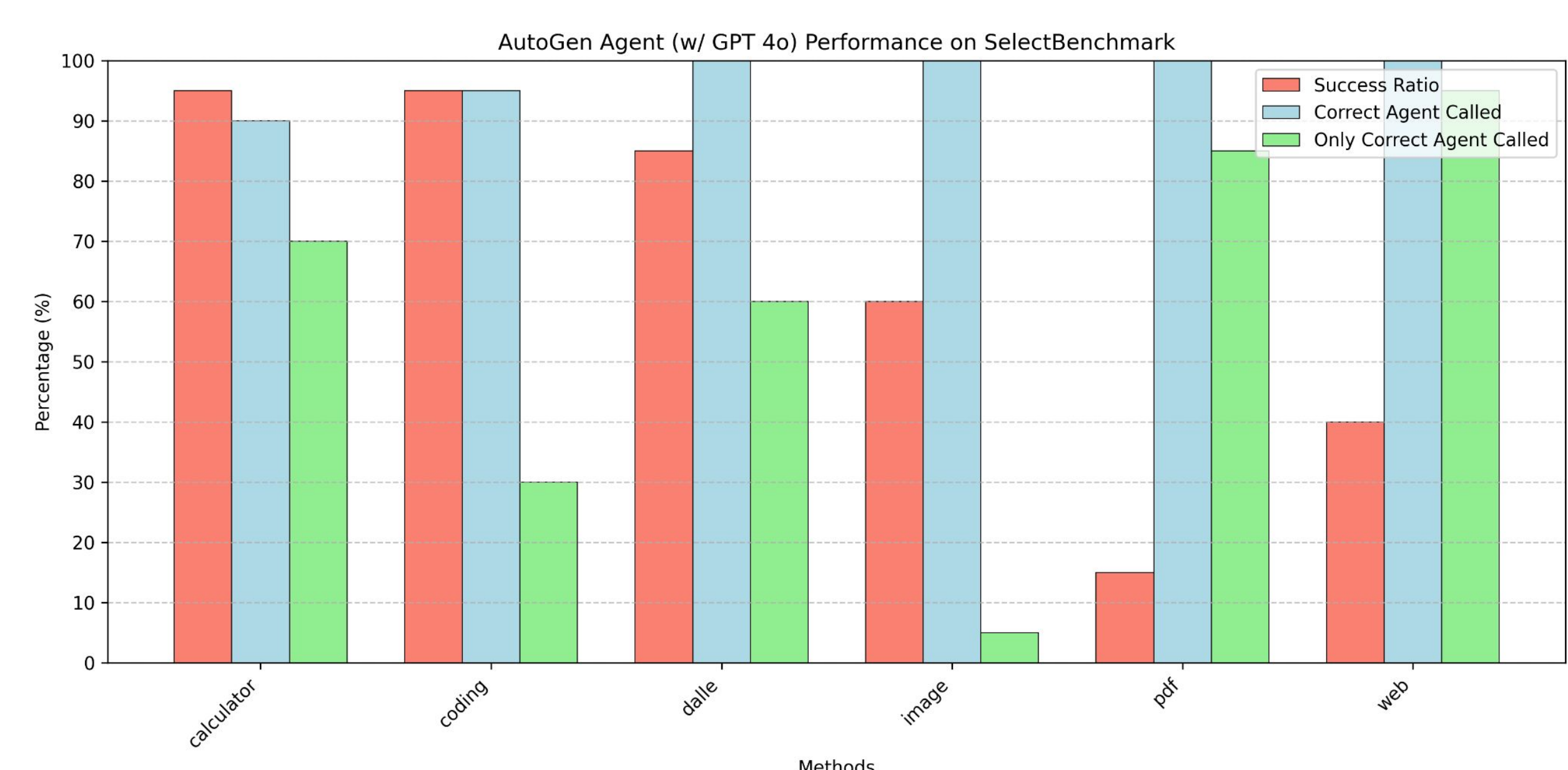
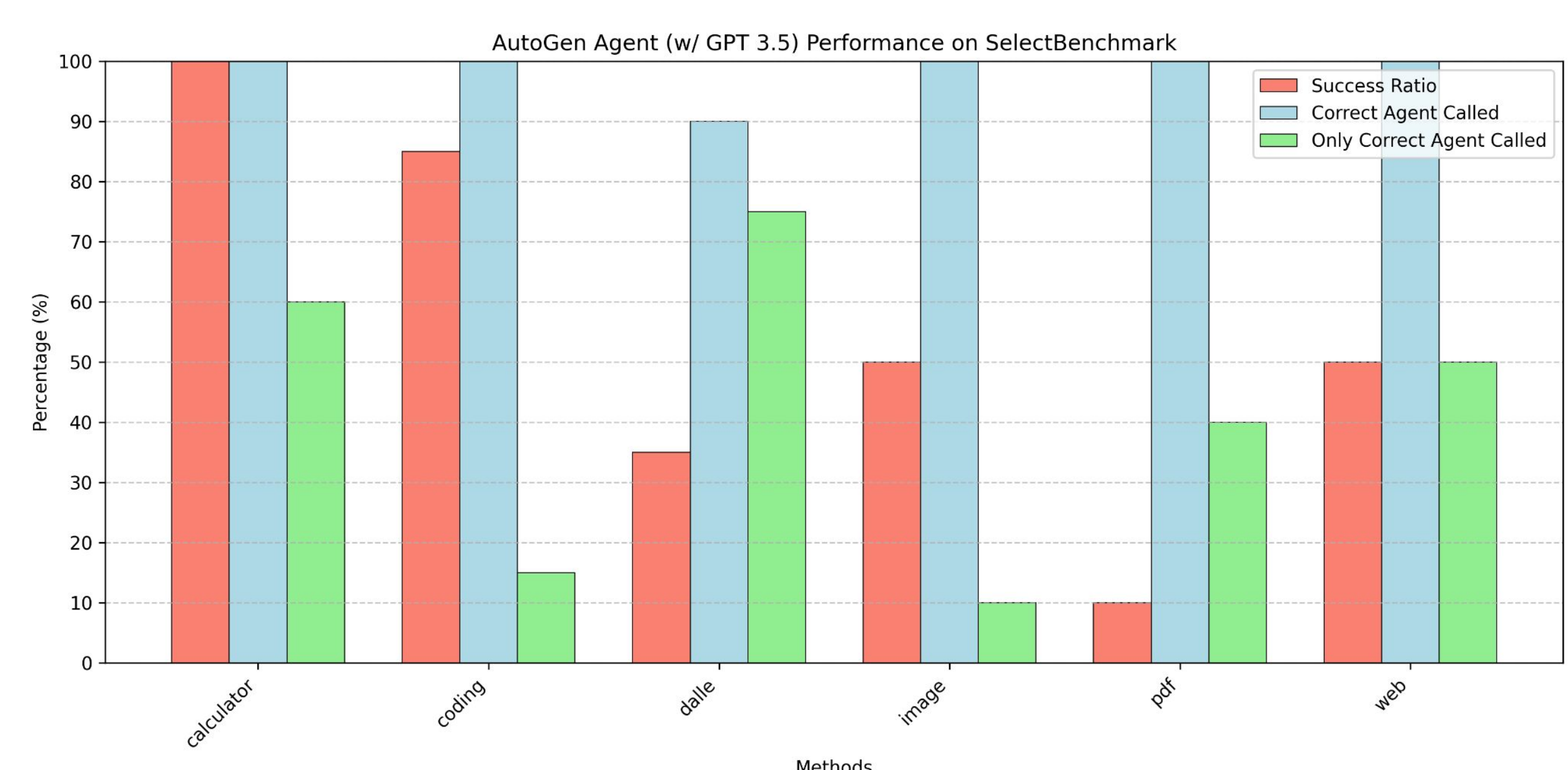


Overarching Agent Chat Example



LLM-Based Evaluation

- ❖ Developed **SelectBenchmark**, a custom benchmark with **120 diverse tasks across 6 categories** with 20 tasks each
- ❖ Given a task, evaluate if the parent agent called
 - **Only Correct Agent** Checks if only the correct child agent is called
 - **Correct Agent:** Checks if the correct child agent is included
 - **Success Ratio:** Measures performance of the called agents on the task
- ❖ Automated evaluation using the LLM-as-evaluator approach for scalable assessment
- ❖ Tested parent and child agents with both **GPT-3.5** and **GPT-4o** backbones to compare performance



Experiments

- ❖ Reproduced the experiments conducted in the original AutoGen paper
 - Some experiments showed similar results, others were more random
- ❖ Additional experiments and scenarios to show the framework's capabilities:
 - **PDFTriage:** retrieving PDF context
 - **DALL-E:**
 - Critic agent improves user's prompt
 - Generator agent to create an image based on the improved prompt
 - **Vision agent** for image context
 - **Browsing agent** leveraging Google Search API and Selenium for direct site retrieval
- ❖ Note: OpenAI models in general demonstrated better multiagent capabilities compared to Google Gemini

Limitations

- ❖ Multiplied price and time: each request generates several responses
- ❖ LLMs are unpredictable - a group of LLMs is even less so
- ❖ Development overhead

Future Work

- ❖ Integrate more specialized agents into the flow
- ❖ Fine tune the agents on task-related datasets to provide better answers
- ❖ Test our parent agent against other similar frameworks